

These problems inevitably turn into interruptions for the designer, and now those interruptions are interfering with the next large batch the designer is supposed to be working on. If the drawings need to be redone, the engineers may become idle while they wait for the rework to be completed. If the designer is not available, the engineers may have to redo the designs themselves. This is why so few products are actually built the way they are designed.

When I work with product managers and designers in companies that use large batches, I often discover that they have to redo their work five or six times for every release. One product manager I worked with was so inundated with interruptions that he took to coming into the office in the middle of the night so that he could work uninterrupted. When I suggested that he try switching the work process from large-batch to single-piece flow, he refused—because that would be inefficient! So strong is the instinct to work in large batches, that even when a large-batch system is malfunctioning, we have a tendency to blame ourselves.

Large batches tend to grow over time. Because moving the batch forward often results in additional work, rework, delays, and interruptions, everyone has an incentive to do work in ever-larger batches, trying to minimize this overhead. This is called the large-batch death spiral because, unlike in manufacturing, there are no physical limits on the maximum size of a batch.⁶ It is possible for batch size to keep growing and growing. Eventually, one batch will become the highest-priority project, a “bet the company” new version of the product, because the company has taken such a long time since the last release. But now the managers are incentivized to increase batch size rather than ship the product. In light of how long the product has been in development, why not fix one more bug or add one more feature? Who really wants to be the manager who risked the success of this huge release by failing to address a potentially critical flaw?

I worked at a company that entered this death spiral. We had been working for months on a new version of a really cool product.